

Numerical Project in Python n.1

Image deblurring

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1 Introduction

This report addresses the signal processing problem of image deblurring. If we consider a one-channel image as a column vector of pixels $z \in \mathbb{R}^n$, and W^H as a transformation matrix such that $z = W^H x$, the observed blurred image b can be modeled as $b = CW^H x$, where C represents the blurring operator.

Reconstructing the original healthy image requires solving an optimization problem.

The problem is thus formulated as the following optimization problem:

$$x^* \in \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} \|CW^H x - b\|_2^2 + \epsilon \|x\|_1 \quad (1)$$

with the final reconstructed image given by $z^* = W^H x^*$.

In this project, we analyze this objective function by splitting it into a smooth part $f_1(x) = \frac{1}{2} \|CW^H x - b\|_2^2$ and a non-smooth part $f_2(x) = \epsilon \|x\|_1$.

We then implement a forward-backward method to solve it, specifically focusing on the Iterative Soft-Thresholding Algorithm (ISTA) and its Nesterov's accelerated version (FISTA).

2 Problem Analysis

2.1 Regularity of the problem

f_1 is a quadratic function of the least-squares form and f_2 is a norm so they are both convex. Finally $f_1 + f_2$ is convex.

f_1 is differentiable (norm squared), convex, et its gradient is L-Lipschitz:

$$\|\nabla f_1(x) - \nabla f_1(y)\| = \|A^T(Ax - b) - A^T(Ay - b)\| \leq L\|x - y\| \quad (2)$$

∇f_1 is L-Lipschitz where L is the largest eigen value of $A^T A$.

2.2 Proximal gradient method

The proximal gradient method minimizes the composite objective $f_1(x) + f_2(x)$ by performing a forward gradient descent step on f_1 followed by a backward proximal step on f_2 . For a step size $\alpha > 0$, the iterations are given by:

$$v_k = x_k - \alpha \nabla f_1(x_k) \text{ and } x_{k+1} = \text{prox}_{\alpha f_2}(v_k) \quad (3)$$

Since $f_2(x) = \epsilon \|x\|_1$, we need to compute the proximal operator for the scaled ℓ_1 norm. As demonstrated in Example 3.3 of the course notes, the proximal operator of the ℓ_1 norm is the soft-thresholding operator. By applying this result with the parameter $\alpha\epsilon$, we directly obtain:

$$[x_{k+1}]_i = \text{sign}([v_k]_i) (|[v_k]_i| - \alpha\epsilon)_+ \quad (4)$$

3 ISTA and FISTA results analysis

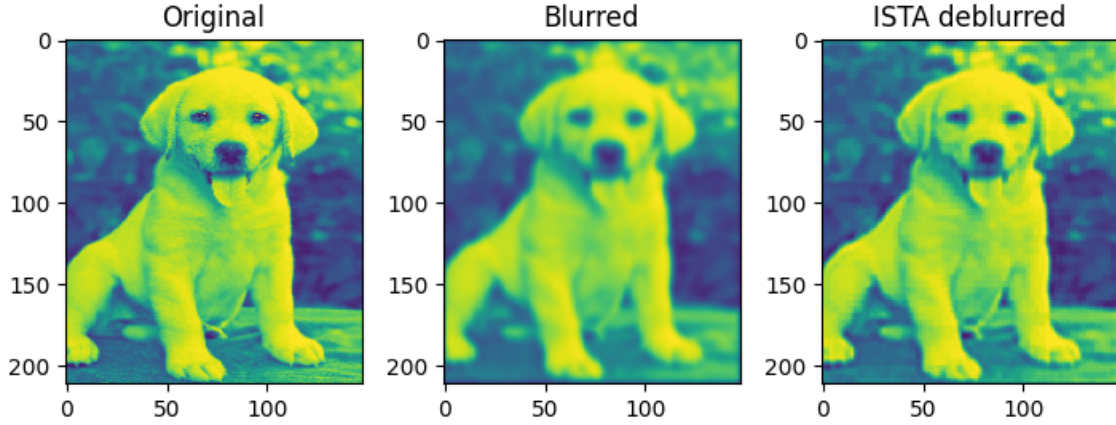


Figure 1: ISTA method

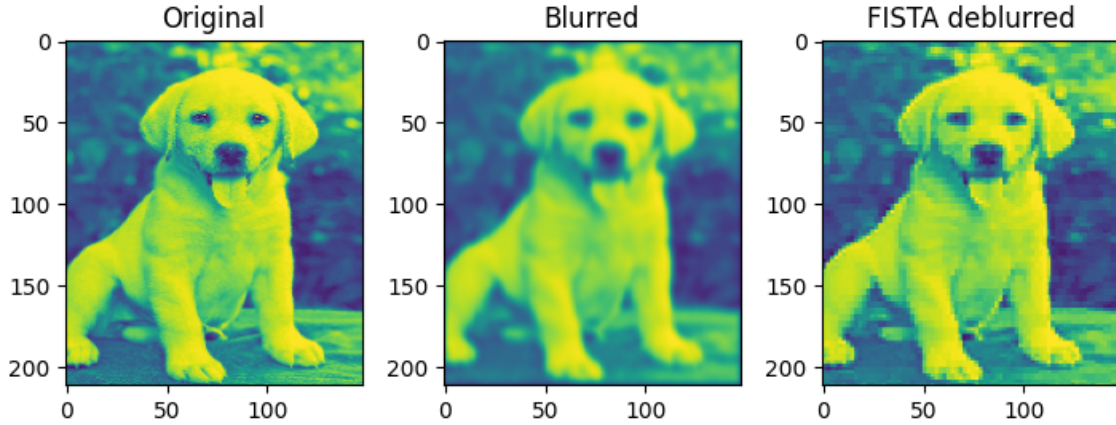


Figure 2: FISTA method

The convergence graph perfectly illustrates the efficiency gap between the standard ISTA algorithm and its accelerated version, FISTA. On a logarithmic (log-log) scale, the optimality gap of the ISTA method decreases steadily and remains parallel to the reference line, which confirms its theoretical sublinear convergence rate of $\mathcal{O}(1/k)$. Conversely, the FISTA curve drops much more rapidly, closely following the slope of the $\mathcal{O}(1/k^2)$ theoretical bound.

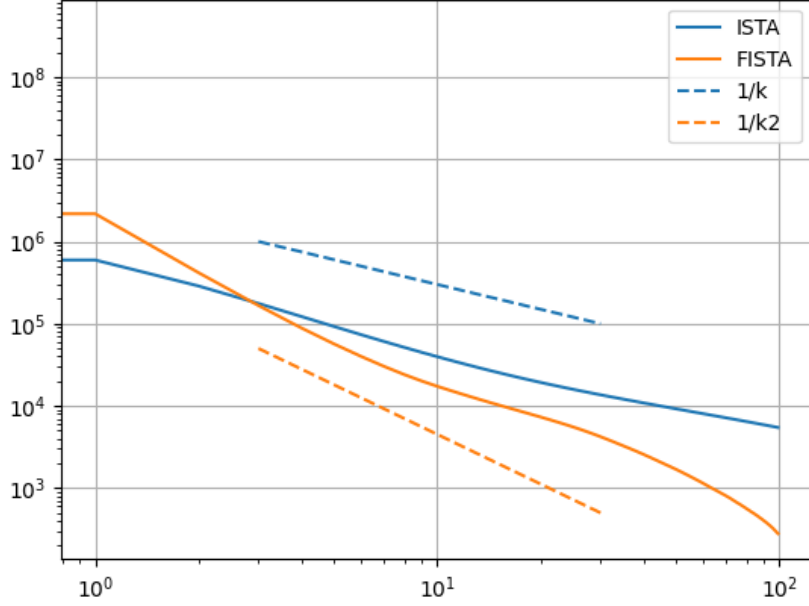


Figure 3: Convergence speed

An interesting visual phenomenon can be observed when comparing the reconstructions: while FISTA successfully removes the blur, the resulting image may appear a little more pixelated than the ISTA result. This is not an anomaly, but rather a direct consequence of FISTA’s superior convergence rate.

Because FISTA efficiently approaches the true mathematical optimum of the objective function, it strictly enforces the ℓ_1 regularization penalty. When transformed back into the pixel domain, this excessive sparsity manifests as more pixelated.

Conversely, because ISTA converges much slower, it has not yet fully enforced this sparsity constraint. The remaining residual blur in the ISTA image acts as a smoothing filter, which paradoxically masks these pixelation artifacts, making the image look softer but mathematically less optimal. This observation perfectly highlights the sensitivity of the regularization parameter ϵ : if the FISTA image looks too pixelated, it implies that ϵ is set slightly too high for this specific image’s texture.

Visually, this algorithmic superiority translates into a significantly better reconstruction quality for FISTA under the exact same computational budget. The image restored by FISTA exhibits much sharper edges and excellent recovery of high frequencies, whereas the image processed by ISTA retains substantial residual blur, proving that the standard algorithm is still far from the optimum.

4 Influence of Optimization Parameters

4.1 The regularization parameter ϵ

The parameter ϵ (initially set to 0.1) is the core of our ℓ_1 regularization. It controls the delicate trade-off between data fidelity (the least-squares term) and the sparsity of the solution.

- **If ϵ is small:** The image quality is higher but if ϵ is too small the ℓ_1 penalty becomes

negligible. The problem reverts to a highly ill-conditioned least-squares inversion.

- **If ϵ is too large:** The solution is forced to be overly sparse. The algorithm over-regularizes the signal, erasing not only the blur but also legitimate image details. This produces an excessive smoothing effect where fine textures disappear entirely.

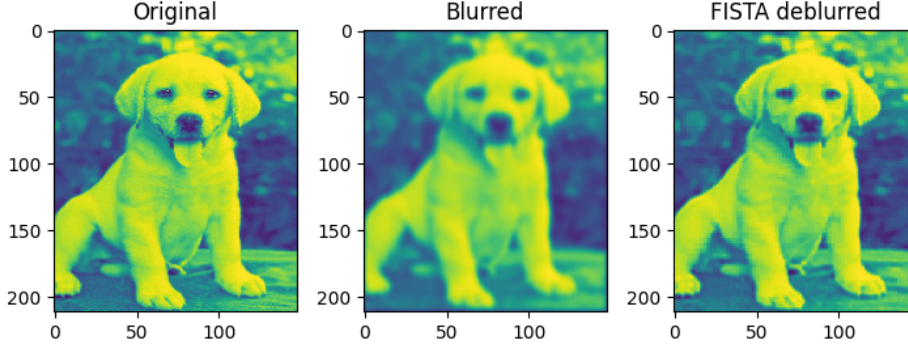


Figure 4: $\epsilon = 0.01$

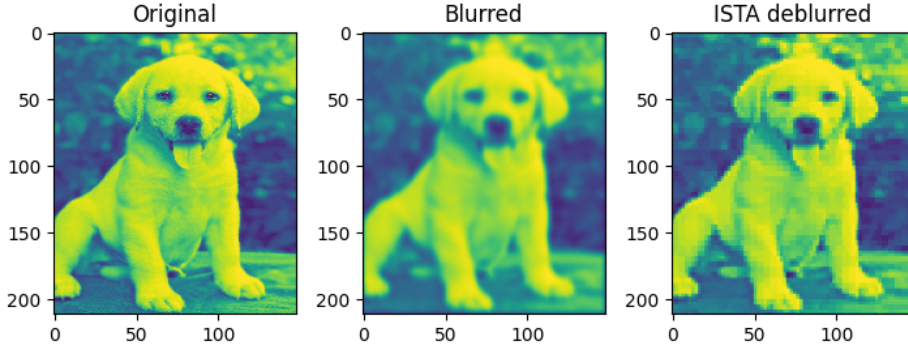


Figure 5: $\epsilon = 0.5$

4.2 The step size α

The descent step size α dictates the stability and speed of the algorithm. Theory requires choosing a step size such that $\alpha \leq 1/L$ to guarantee the convergence of FISTA.

- **Decreasing the step size ($\alpha < 1/L$):** The algorithm remains completely stable but becomes extremely slow. The gradient descent takes very small steps, requiring a significantly higher number of iterations to successfully dissipate the blur. If the number of iterations remains the same, the image appears still blurred.
- **Increasing the step size ($\alpha > 1/L$):** By exceeding the theoretical bound imposed by the Lipschitz constant, the algorithm loses its convergence guarantee. In practice, we rapidly observe a divergence of the iterates, leading to the explosion of the cost function and a totally corrupted image.

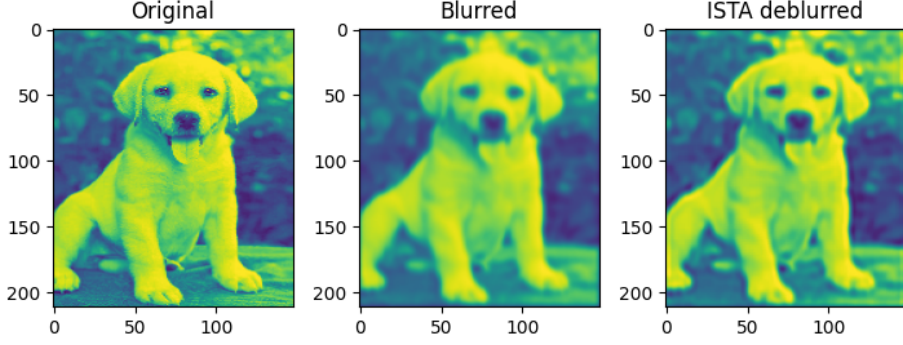


Figure 6: $\alpha = 0.1/L$

4.3 The blurring factor

The blur operator corresponds to a spatial convolution governed by a Gaussian function, whose width is defined by coefficients $(-0.1$ and -0.3 in our base implementation). Modifying these parameters equates to changing the conditioning of the matrix $A = CW^H$.

- **Decreased blur (factors divided by 10):** The convolution kernel becomes highly concentrated and approaches the identity matrix. The inverse problem becomes "easy" and well-conditioned. The algorithms converge in a very small number of iterations to recover a near-perfect image.
- **Increased blur (factors multiplied by 10):** The Gaussian kernel spreads considerably, averaging a large number of neighboring pixels together and irreparably destroying the high frequencies of the signal. Even an optimal algorithm like FISTA struggles to reconstruct fine details, because the original information was physically lost during the observation process.

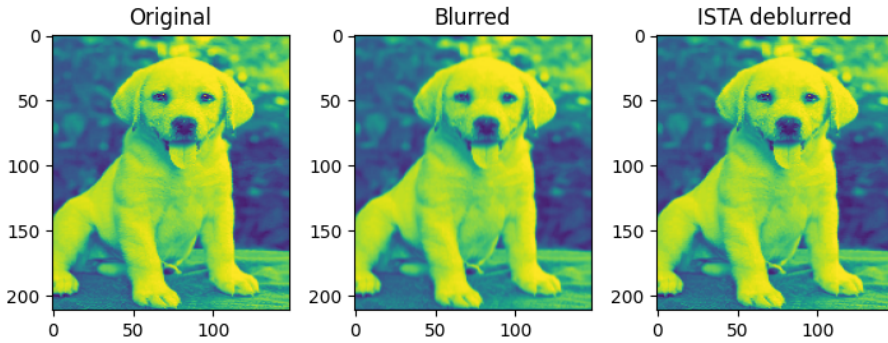


Figure 7: Factors divided by 10

5 Conclusion

In this project, we successfully addressed the image deblurring problem by formulating it as an ℓ_1 regularized least-squares optimization task. By implementing and comparing the standard proximal gradient method (ISTA) and its accelerated variant (FISTA), we empirically validated the profound impact of Nesterov's acceleration.

Our convergence analysis confirmed that FISTA achieves a drastically faster convergence rate of $\mathcal{O}(1/k^2)$, compared to the $\mathcal{O}(1/k)$ rate of ISTA. Furthermore, our parametric study highlighted the critical importance of optimization parameters: the step size α must be strictly bounded to ensure algorithmic stability, while the regularization parameter ϵ must be carefully balanced to avoid excessive sparsity and pixelation artifacts.

Overall, this project practically demonstrated the power and necessity of accelerated forward-backward splitting methods for solving large-scale, non-smooth inverse problems in image processing.